

other harmful metals.

Recycling and disposal of e-waste may pose significant risk to workers and communities. Therefore great care must be taken to avoid unsafe exposure in recycling operations and leaching of materials such as heavy metals from landfills and incinerator ashes. As such, lead is not a problem when contained in electronic equipment, but when electronic components are deposited in landfills, people may scavenge and break these open or the lead may leach out of landfills and into drinking water. The risk compounds in countries that are massive importers of electronic waste.

Though no study demonstrates environmental or human health risk posed by electronic products in landfills, electronic waste disposal through recycling process poses a great threat to its workers and the environment. Lead is not absorbed through the skin and exposure is through inhalation or ingestion.

The main impetus for the industry to leave lead behind is a ban on lead in electronics imposed in many developed countries. Accordingly, lead must be replaced with other substances in electronic equipment.

Lead-free alternatives for solder

Solder is the linchpin of electronics manufacturing without which it's difficult to achieve a proper electronic connection that is durable and reliable. Lead has been ideal for solder. The electronics industry is learning to do without lead, for which it has to abandon lead-tin solder—one of its long-time staples.

For decades lead-tin solder has been used to attach electronic components to printed wiring boards. However, with the body of evidence pointing to adverse health effects of lead, the search for a replacement has spawned intense efforts in the electronics industry and in universities.

Now scientists think they have found some promising leads: solders



Lead electronics



E-waste



E-waste reclamation



Lead poisoning

made of alternative alloys and polymer formulations known as electrically conductive adhesives. Even as efforts to replace lead in solder move ahead, there are still concerns about the im-

pact that newly implemented metals will have on human and environmental health. The alternatives to lead have not been researched extensively in terms of potential health and environmental impacts. Getting rid of lead doesn't mean to replace it with something that is obviously safer, but to keep looking for lead alternatives that are environmentally benign. Indeed, no metallic alternative to lead is free from environmental concerns.

Solders made of alternative alloys

Because of the considerable toxicity of lead, efforts are being made to replace the traditional soldering alloys with new compositions. Research aims to improve the reliability of lead-free soldering alloys that are used to make electronic devices. However, the reliability of new lead-free materials requires further investigation.

The main approach to replacing lead in solder has been to look for other metals as substitutes. Electronics manufacturers began to look for alternative metals in the 1990s. A search by industry experts for possible replacements for lead-tin solder winnowed down 75 metal alloy alternatives to about half a dozen. The biggest benefit for the industry would be to pick one solder, concentrating the development and research efforts on one alloy and making it work. The industry eventually selected a tin-silver-copper combination as the most reliable and easy-to-work-with replacement. The formulation—95.5 per cent tin, 3.9 per cent silver, 0.6 per cent copper—is also known as SAC solder, for the first letters of the chemical symbols of each of the elements (Sn, Ag, Cu).

Tin-silver-copper appears to have reliability at least as good as, if not higher than, tin-lead. With a melting point of 217°C, SAC solder is also the closest in melting point to the conventional lead-tin solder. This does mean, however, an unquantified increase in energy use. Furthermore, the higher temperature may pose problems for